b. Tristyly: The plant produces three kinds of flowers, with respect to the length of the style and stamens. Here,the pollen from flowers of one type can pollinate only the other two types but not their own type. Example : *Lythrum* (Figure 1.14b).

iv. Self sterility/ Self- incompatibility: In some plants, when the pollen grain of a flower reaches the stigma of the same, it is unable to germinate or prevented to germinate on its own stigma. Examples: *Abutilon*, *Passiflora*. It is a genetic mechanism.

Agents of pollination

Pollination is effected by many agents like wind, water, insects etc. On the basis of the agents that bring about pollination, the mode of pollination is divided into abiotic and biotic. The latter type is used by majority of plants.

Abiotic agents

1. Anemophily - pollination by Wind
2. Hydrophily - pollination by Water

Biotic agents

3. Zoophily

Zoophily refers to pollination through animals and pollination through insects is called Entomophily.

1. Anemophily: Pollination by wind. The wind pollinated flowers are called **anemophilous**. The wind pollinated plants are generally situated in wind exposed regions. Anemophily is a chance event. Therefore, the pollen may not reach the target flower effectively and are wasted during the transit from one flower to another. The common examples of wind pollinated flowers are - grasses, sugarcane, bamboo, coconut, palm, maize etc.,

Anemophilous plants have the following characteristic features:

- The flowers are produced in pendulous, catkin-like or spike inflorescence.
- The axis of inflorescence elongates so that the flowers are brought well above the leaves.
- The perianth is absent or highly reduced.
- The flowers are small, inconspicuous, colourless, not scented, do not secrete nectar.
- The stamens are numerous, filaments are long, exerted and versatile.
- Anthers produce enormous quantities of pollen grains compared to number of ovules available for pollination. They are minute, light and dry so that they can be carried to long distances by wind.
- In some plants anthers burst violently and release the pollen into the air. Example: *Urtica.*
- Stigmas are comparatively large, protruding, sometimes branched and feathery, adapted to catch the pollen grains. Generally single ovule is present.
- Plant produces flowers before the new leaves appear, so the pollen can be carried without hindrance of leaves.

Pollination in Maize (*Zea mays*): The maize is monoecious and unisexual. The male inflorescence (tassel) is borne terminally and female inflorescence (cob) laterally at lower levels. Maize pollens are large and heavy and cannot be carried by light breeze. However, the mild wind shakes the male inflorescence to release the pollen which falls vertically below. The female inflorescence has long stigma (silk) measuring upto 23 cm in length, which projects beyond leaves. The pollens drop from the tassel is caught by the stigma (Figure 1.15).

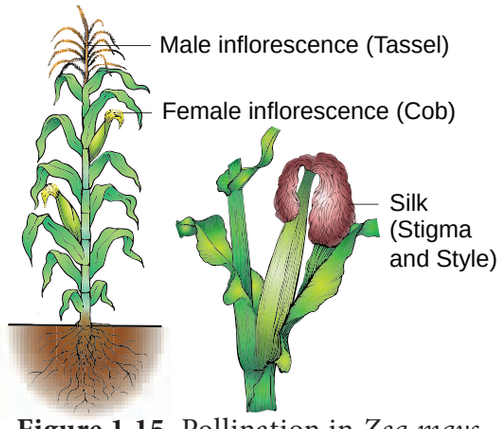


Figure 1.15 Pollination in *Zea mays*

TN_GOVT_BIOLOGY-BOTANY_XII_PAGES_001-032 CH 01.indd 15 3/14/2020 10:30:05 AM